

Minor Project

Exploratory Data Analysis (EDA) Project

Student Project Guidelines | Jupyter Notebook Submission

Project Objective

You will receive a real-world dataset from Kaggle. Your task is to understand, clean, analyze, and visualize the dataset using Python. The completed project must be submitted as a Jupyter Notebook (.ipynb).

1. Project Introduction

- Project title
- Student name
- Dataset name
- Kaggle dataset link
- Short description of the dataset
- Objective of the analysis

2. Load the Dataset

- Import the required libraries.
- Load the dataset.
- Display the first and last few records.



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3. Understand the Dataset

- Examine the number of rows and columns.
- Review column names and data types.
- Check the dataset information and statistical summary.
- Examine unique values in each column.
- Write a short observation about the dataset.

4. Check and Remove Duplicate Records

- Check for duplicate rows.
- Remove duplicates if present.
- Mention the number of duplicates found.
- Compare the dataset size before and after removing duplicates.

5. Handle Missing Values

- Check missing values in every column.
- Choose a suitable method to handle them.
- Confirm that missing values have been handled.
- Explain the reason for choosing that method.

6. Clean the Dataset

- Correct incorrect data types.
- Remove unwanted symbols and extra spaces.
- Standardize different spellings of the same category.
- Handle invalid or unknown values.
- Remove unnecessary columns only when appropriate.



7. Analyze Categorical Columns

- Check unique categories and category counts.
- Create appropriate graphs.
- Write observations below the graphs.

8. Analyze Numerical Columns

- Examine mean, median, minimum, maximum, and standard deviation.
- Study the distribution of values.
- Create appropriate graphs and write observations.

9. Detect and Handle Outliers

- Use suitable visualizations to identify outliers.
- Handle outliers only when necessary.
- Explain the method used.
- Compare the data before and after handling outliers.
- Do not remove outliers without giving a proper reason.

10. Perform Univariate Analysis

- Analyze one variable at a time.
- Study numerical distributions, category frequency, common values, and minimum and maximum values.

11. Perform Bivariate Analysis

- Analyze the relationship between two variables.
- Include numerical-to-numerical and numerical-to-categorical comparisons where appropriate.



12. Perform Multivariate Analysis

- Analyze relationships among multiple variables.
- Include correlation analysis, a correlation heatmap, and other suitable visualizations.

13. Write Key Insights

- Write at least 5-10 meaningful insights based on your analysis.
- Do not simply describe a graph. Explain what the result tells us about the dataset.

14. Final Conclusion

- Summarize what you learned from the dataset.
- Mention important patterns discovered.
- Discuss major data-quality problems found.
- Explain how the dataset could be useful for a future machine-learning project.

Notebook Presentation Rules

- Use both Markdown and code cells.
- Add clear headings for every section.
- Keep the work simple, clean, and readable.
- Display all important outputs.
- Add titles and labels to every graph.
- Write observations below important outputs and graphs.
- Ensure the notebook runs from the first cell to the last cell without errors.
- Avoid unnecessary or repeated work.
- End with one final cleaned DataFrame, preferably named `df_clean`.



Submission Requirements

- Completed .ipynb notebook
- Kaggle dataset link
- Cleaned .csv dataset, if requested
- GitHub repository link, if requested

File-Naming Format

Example: Anjali_UsedCars_EDA_Project.ipynb

Important Instructions

- The submitted notebook must be your own work.
- Do not copy another student's notebook.
- Do not submit AI-generated work without understanding it.
- Be prepared to explain your cleaning decisions, analysis, graphs, observations, and conclusion.
- Late or incomplete submissions may not be accepted.



Submission

Deadline:

___ Mentor: Sruthi

Tarimana



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